

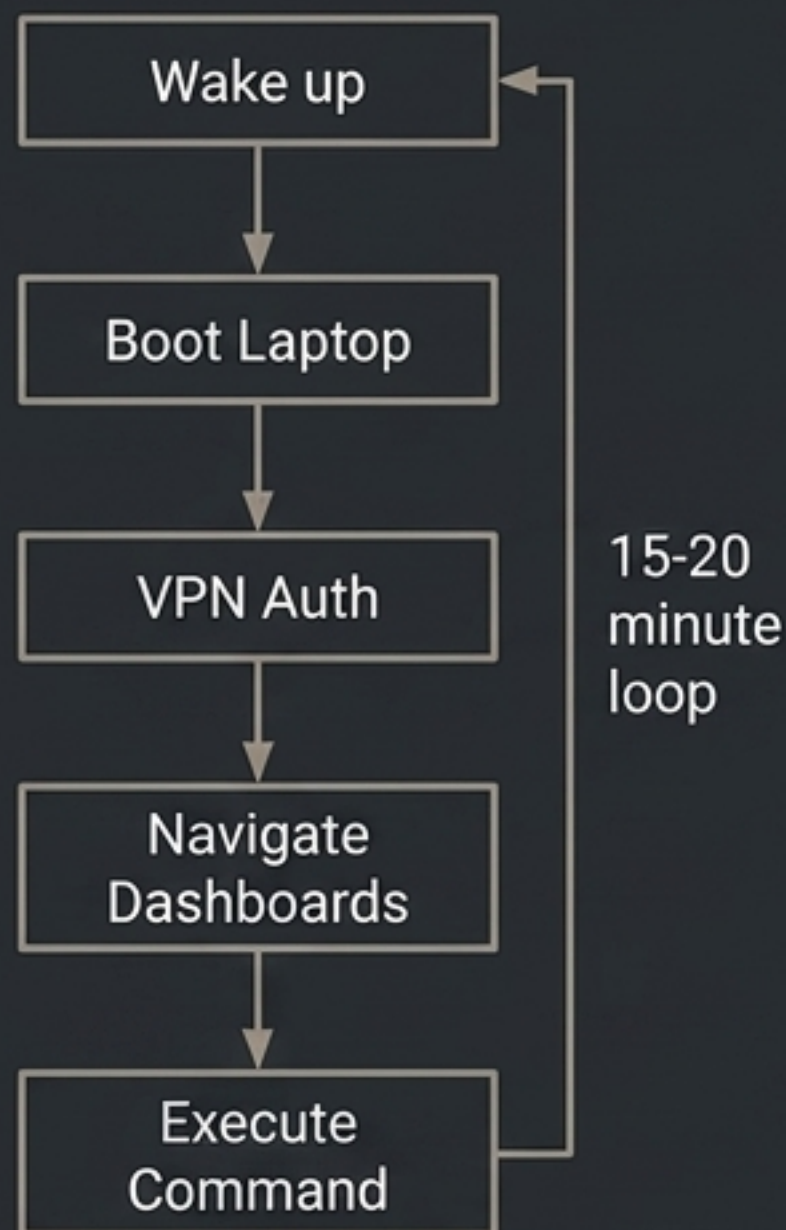
CALL-E HACKATHON SUBMISSION 

PagerZero: Zero-Wake On-Call

Autonomous AI SRE pager replacement that **resolves** routine outages **silently** and secures voice approval over the phone for critical **fixes** while engineers stay in bed.

The 3:00 AM Alert Fatigue Crisis

Status Quo Friction



80%+ of on-call pages are routine incidents with well-defined runbooks (log disk cleanup, stale cache, pool restarts).

Waking engineers causes chronic sleep disruption, cognitive degradation, and rapid engineering turnover.

Laptops take 15-20 minutes to open, authenticate via VPN, navigate dashboards, and execute single-command remediations.

The PagerZero Mission



Enable autonomous resolution with spoken voice approvals only when human consent is required.

The Autonomous Escalation Funnel

88%

Fewer Middle-of-Night Awakenings.

✓ Tier 1: Auto-Fix

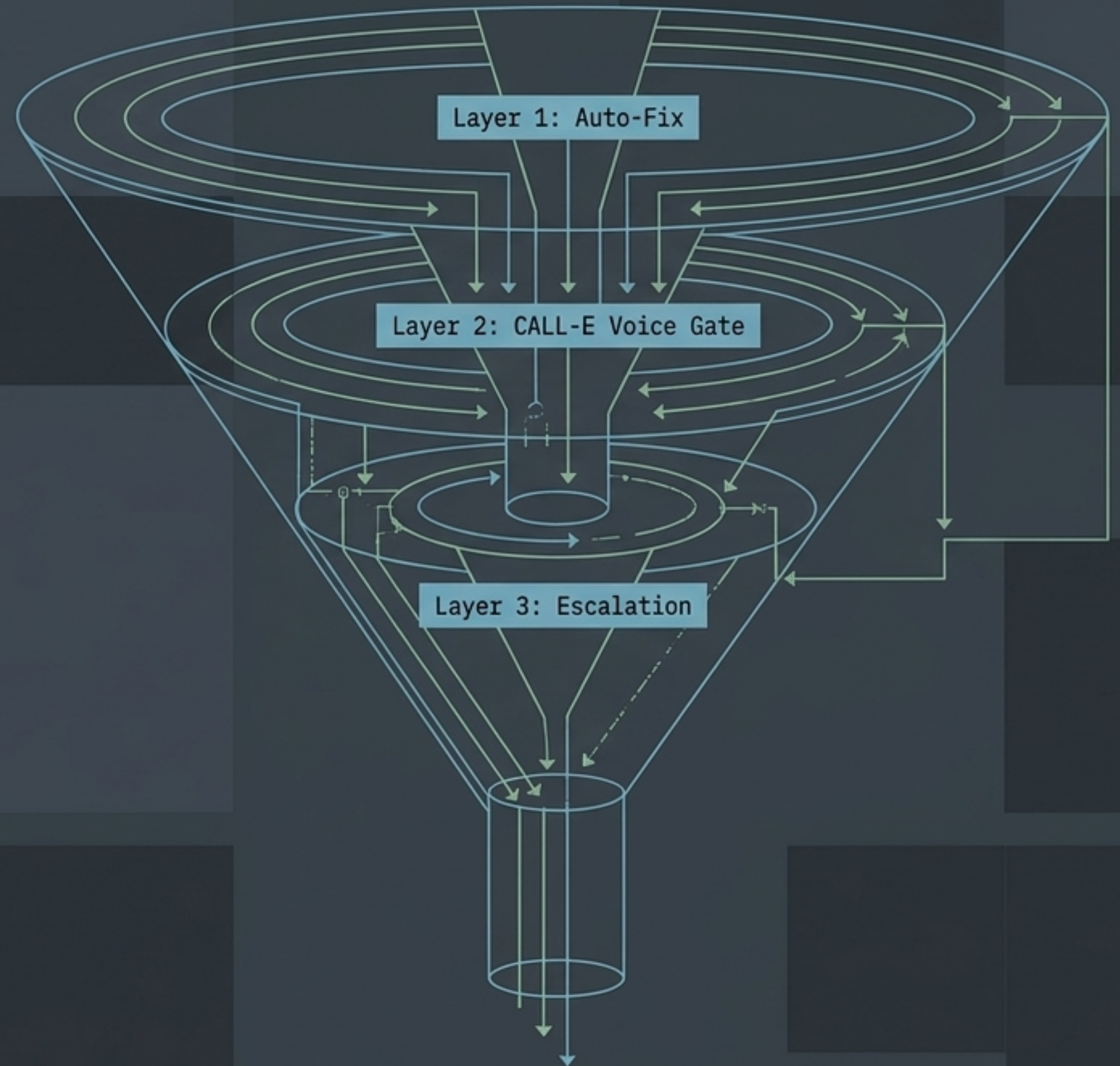
Low-risk, idempotent runbooks (log pruning, cache flush) execute autonomously in milliseconds. Telemetry verifies health and post-mortems emit silently.

📞 Tier 2: CALL-E Voice Gate

High-impact actions (DB pool recycle, pod rolling restart) trigger a 15-second phone call. Engineer speaks "Approved" and goes back to sleep.

⚠️ Tier 3: Escalation

If rejected, call drops, or voice timeout elapses, the incident immediately escalates to secondary on-call rosters.



Separation of Concerns: Reasoning vs. Voice

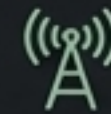


SRE Reasoning Engine

CPU_LOAD: 85% | LATENCY: 120ms | ERROR_RATE: 0.02% | CONN_STATE: ESTABLISHED

Responsibility: Deep telemetry parsing, stack trace evaluation, runbook candidate ranking, and risk assessment.

Mechanism: Ingests real-time metrics across CPU, latency, error rates, and connection states to formulate targeted remediation commands before placing calls.



CALL-E Telephony Gateway

WAVEFORM: [|||||] SIGNAL_STRENGTH: HIGH

Responsibility: Cellular network dialing, conversational audio turn-taking, and structured JSON schema extraction.

Mechanism: Replaces brittle IVR menus with natural dialogue, handles background noise and interruptions, and returns a verified decision payload.



Zero Laptops. One Spoken Word.

PagerZero bridges SRE reasoning directly to mobile telephony via the CALL-E SDK.

The Interaction: In under 15 seconds, the engineer hears the root cause, agrees to the remediation, and hangs up.

The Audit Trail: Structured schema extraction confirms spoken intent with complete cryptographic logging and auditability.

```
{
  "error": "DatabaseConnectionLost",
  "stack": [
    "at Connection.connect (./db.js:45:12)",
    "at Request.execute (./api.js:102:5)",
    "at ...",
    ...
  ],
  "timestamp": "1679483198"
}
```



```
{
  "status": "approved",
  "timestamp": "1679483201",
  "signature": "0x4f8..."
}
```



The Autonomous Execution Loop

Human Response Delay: 28.0 mins

Wake up

Find laptop

VPN login

0.0s: Anomaly
Detected

1.5s: Policy
Evaluation

6.0s: Spoken
Approval

8.0s: Recovery
Verified

Prometheus or Datadog
webhook ingested. SRE
diagnostic agent
evaluates telemetry.

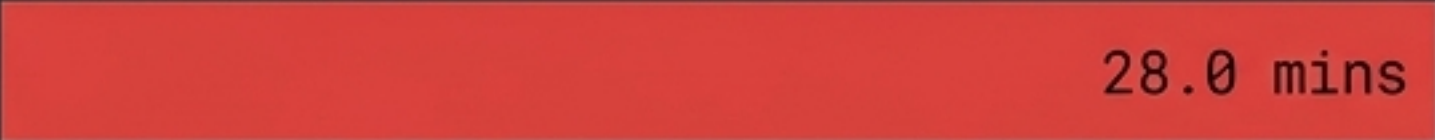
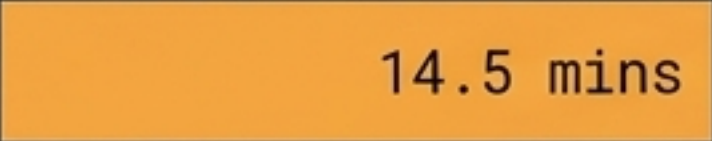
Risk engine evaluates
impact. Tier 1 auto-
resolves; Tier 2 triggers
CALL-E outbound call.

Engineer answers
phone, hears 15s
briefing, and speaks
"Approved".

Remediation executes.
Telemetry verified
healthy and post-
mortem published.

Mean Time to Recovery Comparison

PagerZero slashes MTTR by up to 99% compared to traditional workflows by removing human delay from diagnostic and execution loops.

Workflow	Mean Time to Recovery	Cognitive Disruption	Tooling Required
Traditional Pager + Laptop	 28.0 mins	High	Laptop + VPN Required
ChatOps Slack Notification	 14.5 mins	Medium	Mobile App/Laptop
PagerZero Tier 2 (Voice)	 1.2 mins	Near-Zero	Cellular Audio Only
PagerZero Tier 1 (Auto-Fix)	 0.15 mins	Zero	Fully Autonomous

| Production-Grade Safety Guardrails



Pre-Execution State Validation: Assesses blast radius and verifies service dependencies prior to triggering any runbook.



Idempotency Guarantees: Every remediation script is built to be safely retried without unintended state mutation or data loss.

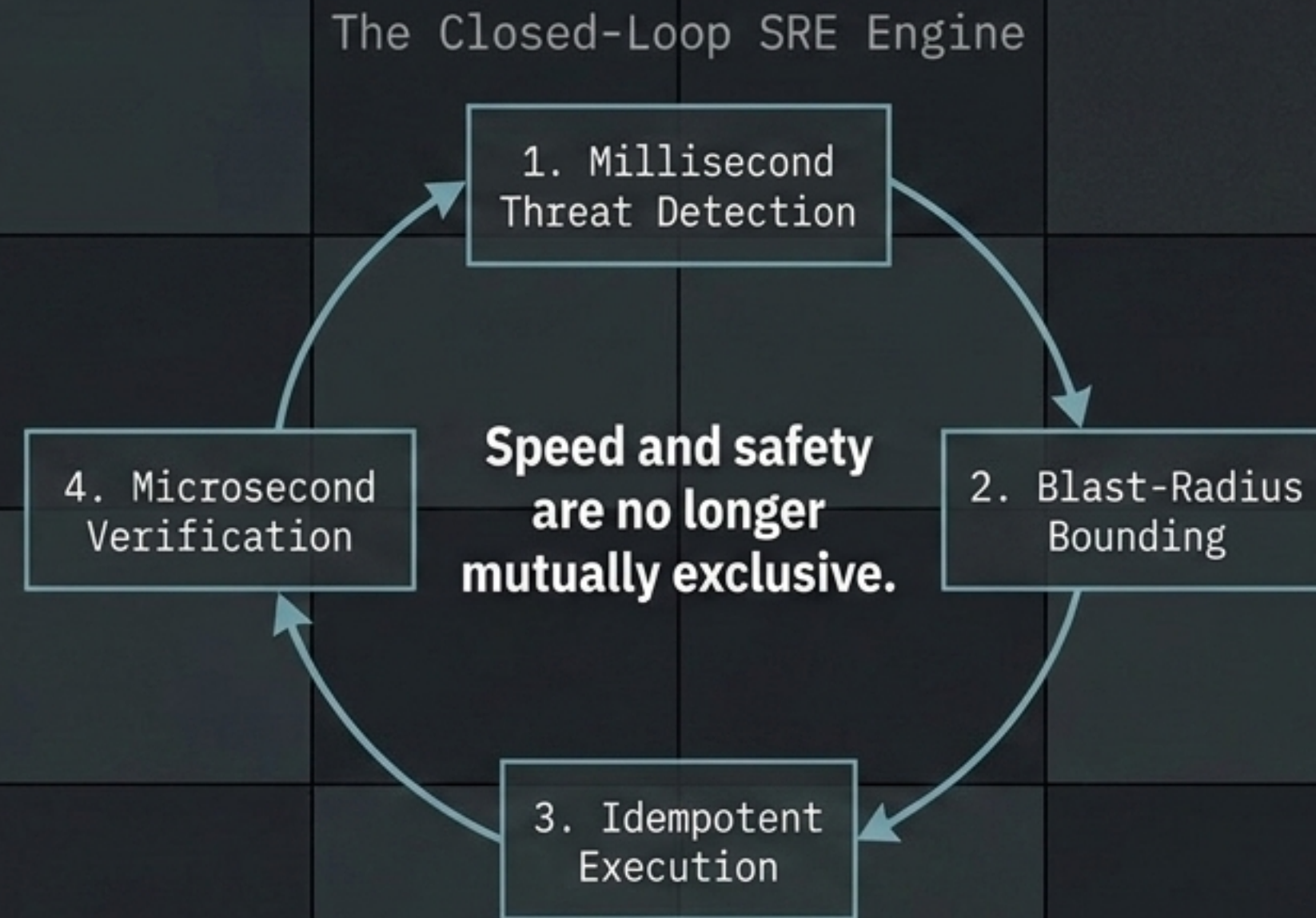


Automated Rollback Triggers: If latency or error rate fails to normalize within 5 seconds of remediation, changes rollback instantly.



Instant Post-Mortem Generation: Generates markdown RCA reports with timeline telemetry, spoken transcripts, and audit IDs.

| The Safety-Speed Synthesis



By executing idempotently and auto-rolling back in 5 seconds, autonomous mitigation is mathematically safer than a 20-minute human response window where cascading failures propagate.

Chaos Outage Verification Matrix

Failure Scenario	Telemetry Symptom	Resolution Policy	Action Taken	Verification Status
Disk Volume Saturation	96.4% disk usage on /data	Tier 1: Autonomous	Rotated logs & pruned caches	✓ 28.4% usage (Restored)
DB Pool Starvation	198/200 pooled connections	Tier 2: CALL-E Voice	Flushed idle client sockets	✓ 38 connections (Restored)
Pod Memory Leaking	OOM kills recurring every 90s	Tier 2: CALL-E Voice	Rolling restart with limit bump	✓ RSS flat at 210MB
API Latency Degradation	p99 latency surged to 4650ms	Tier 1: Autonomous	Tripped circuit breaker to cache	✓ Latency normalized to 24ms

Built for Mission-Critical Teams

PagerZero combines the diagnostic precision of large reasoning models with the cellular immediacy of CALL-E to protect your production reliability around the clock.

Sleep Through Routine Outages. Experience the next generation of voice-approved autonomous SRE operations.



pagerzero.pro